

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – VII

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch: 1
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AIM : Sampling and Sampling Distributions

A) Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.

1. Select a real-life business topic for conducting a survey.

Topic : Customer Satisfaction with Online Food Delivery Services

2. Prepare a questionnaire containing at least 5 questions related to the selected topic.

Q1) What is your age?

Q2) Which online food delivery platform do you use most?

Q3) How often do you order food online per month?

Q4) What is your average spending per order (₹)?

Q5) How satisfied are you with the food delivery service? (1–5)

Q6) Would you recommend the service to others? (Yes/No)

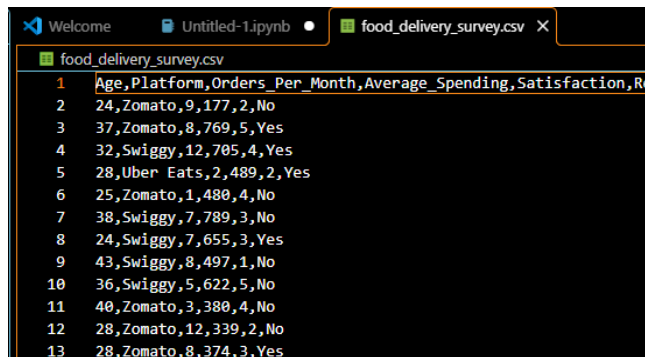
3. Collect responses from at least 50 people.

```
import pandas as pd
import numpy as np
# For reproducible results
np.random.seed(42)
# Number of respondents
n = 50
# Create survey data
data = {
    "Age": np.random.randint(18, 46, n),
    "Platform": np.random.choice(
        ["Swiggy", "Zomato", "Uber Eats"],
        n,
        p=[0.45, 0.45, 0.10]
    ),
    "Orders_Per_Month": np.random.randint(1, 13, n),
    "Average_Spending": np.random.randint(150, 801, n),
    "Satisfaction": np.random.randint(1, 6, n),
    "Recommend": np.random.choice(
```

```

        ["Yes", "No"],
        n,
        p=[0.75, 0.25]
    )
}
# Create DataFrame
df = pd.DataFrame(data)
# Display first 5 records
print("First Five Records:")
print(df.head())
4. Store the collected data in a CSV file or Pandas DataFrame.\
df.to_csv("food_delivery_survey.csv", index=False)
print("CSV file created successfully!")

```



	Age	Platform	Orders_Per_Month	Average_Spending	Satisfaction	Recommend
1	24	Zomato	9	177	2	No
2	37	Zomato	8	769	5	Yes
3	32	Swiggy	12	705	4	Yes
4	28	Uber Eats	2	489	2	Yes
5	25	Zomato	1	480	4	No
6	38	Swiggy	7	789	3	No
7	24	Swiggy	7	655	3	Yes
8	43	Swiggy	8	497	1	No
9	36	Swiggy	5	622	5	No
10	40	Zomato	3	380	4	No
11	28	Zomato	12	339	2	No
12	28	Zomato	8	374	3	Yes

5. Display the first five records of the collected data.

First Five Records:

	Age	Platform	Orders_Per_Month	Average_Spending	Satisfaction	Recommend
0	24	Zomato	9	177	2	No
1	37	Zomato	8	769	5	Yes
2	32	Swiggy	12	705	4	Yes
3	28	Uber Eats	2	489	2	Yes
4	25	Zomato	1	480	4	No

6. Identify the population and sample for the survey.

Population: All customers who use online food delivery services.

Sample: The 50 customers selected for this survey.

7. Identify the sampling technique used to select the sample.

Convenience Sampling

The respondents are selected based on their availability and willingness to participate in the survey.

8. Identify the numerical and categorical variables in the collected data.

Numerical Variables

- Age
- Orders_Per_Month

- Average_Spending
- Satisfaction

Categorical Variables

- Platform
- Recommend

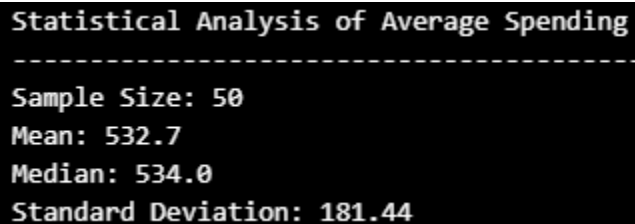
9. Select one numerical variable for further analysis.

10. Calculate the sample size, mean, median, and standard deviation of the selected variable.

CODE :-

```
variable = df["Average_Spending"]
sample_size = variable.count()
mean = variable.mean()
median = variable.median()
std_dev = variable.std()
print("Statistical Analysis of Average Spending")
print("-----")
print("Sample Size:", sample_size)
print("Mean:", round(mean, 2))
print("Median:", round(median, 2))
print("Standard Deviation:", round(std_dev, 2))
```

OUTPUT :-

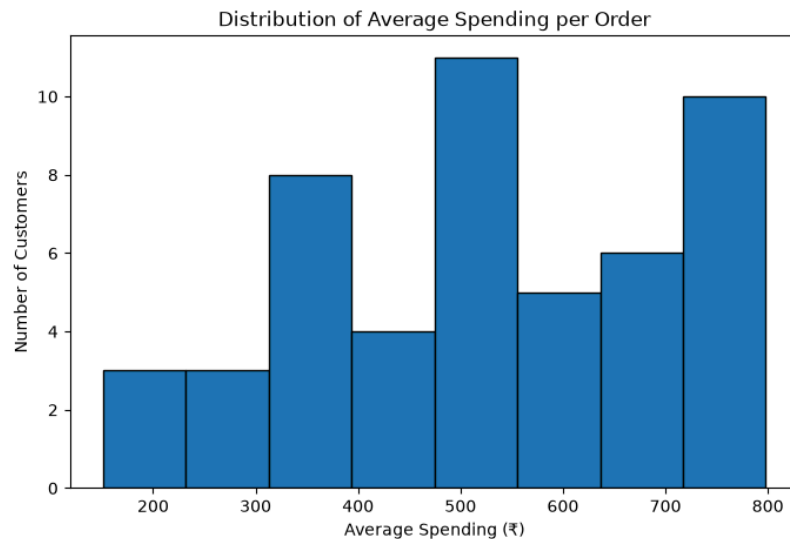


```
Statistical Analysis of Average Spending
-----
Sample Size: 50
Mean: 532.7
Median: 534.0
Standard Deviation: 181.44
```

11. Create a suitable chart to display the collected data.

CODE :-

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 5))
plt.hist(
    df["Average_Spending"],
    bins=8,
    edgecolor="black"
)
plt.title("Distribution of Average Spending per Order")
plt.xlabel("Average Spending (₹)")
plt.ylabel("Number of Customers")
plt.show()
```

OUTPUT :-**12. Write two observations from the survey data.**

- 1) The average spending per order shows how much customers typically spend on online food delivery, with some variation between customers.
- 2) Swiggy and Zomato account for most of the responses, indicating that they are the most commonly used platforms among the surveyed customers.

B) Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

Select a numerical variable from the collected data.

1. Select a sample of 30 observations randomly from the data.**CODE :-**

```
sample = df["Average_Spending"].sample(n=30, random_state=42)
print("Random Sample of 30 Observations:")
print(sample)
```

OUTPUT :-

```
Random Sample of 30 Observations:
13    526
39    774
30    646
45    715
17    408
48    552
```

2. Calculate the mean of the selected sample.**CODE :-**

```
sample_mean = sample.mean()
print("Mean of the selected sample:", round(sample_mean, 2))
```

OUTPUT :-

```
Mean of the selected sample: 545.87
```

3. Repeat the random sampling process 100 times.

CODE :-

```
sample_means = []
for i in range(100):
    sample = df["Average_Spending"].sample(n=30)
    sample_mean = sample.mean()
    sample_means.append(sample_mean)
print("Sampling completed 100 times.")
```

OUTPUT :-

```
Sampling completed 100 times.
```

4. Calculate and store the mean of each sample.

CODE :-

```
sample_means = []
for i in range(100):
    sample = df["Average_Spending"].sample(n=30)
    mean = sample.mean()
    sample_means.append(mean)
print("Sample means:")
print(sample_means)
```

OUTPUT :-

```
Sample means:
[np.float64(523.4), np.float64(500.3), np.float64(540.4333333333333), np.float64(515.8333333333334), np.float64(53
```

5. Display the first 10 sample means.

CODE :-

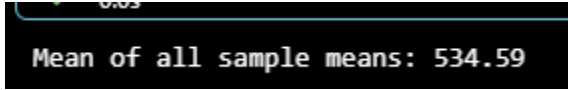
```
print("First 10 Sample Means:")
for i in range(10):
    print(round(sample_means[i], 2))
```

OUTPUT :-

```
First 10 Sample Means:
523.4
500.3
540.43
515.83
537.2
514.2
528.73
552.3
511.3
560.57
```

6. Calculate the mean of all the sample means.**CODE :-**

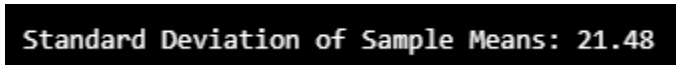
```
mean_of_sample_means = sum(sample_means) / len(sample_means)
print("Mean of all sample means:",
      round(mean_of_sample_means, 2))
```

OUTPUT :-

```
Mean of all sample means: 534.59
```

7. Calculate the standard deviation of the sample means.**CODE :-**

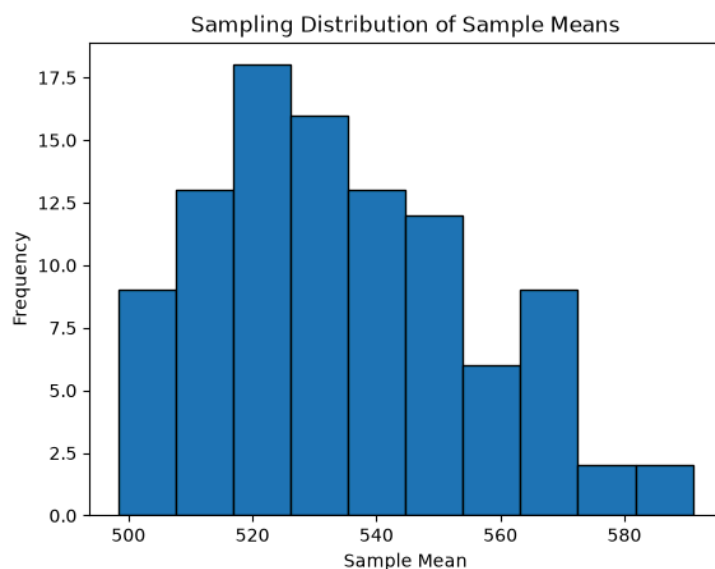
```
import numpy as np
std_sample_means = np.std(sample_means, ddof=1)
print("Standard Deviation of Sample Means:",
      round(std_sample_means, 2))
```

OUTPUT :-

```
Standard Deviation of Sample Means: 21.48
```

8. Create a histogram of the sample means.**CODE :-**

```
import matplotlib.pyplot as plt
plt.hist(sample_means, bins=10, edgecolor="black")
plt.title("Sampling Distribution of Sample Means")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT :-

9. Observe the shape of the sampling distribution.

The sampling distribution is approximately bell-shaped and centered around the mean of the sample means.

10. Compare the original sample mean with the mean of the sampling distribution.**CODE :-**

```
original_mean = df["Average_Spending"].mean()
print("Original Mean:", round(original_mean, 2))
print("Mean of Sampling Distribution:",
      round(mean_of_sample_means, 2))
```

OUTPUT :-

```
Original Mean: 532.7
Mean of Sampling Distribution: 534.59
```

The original sample mean and the mean of the sampling distribution are close to each other.

11. Compare the sampling distributions obtained for different sample sizes.**CODE :-**

```
sample_sizes = [10, 20, 30, 40]
for size in sample_sizes:
    means = []
    for i in range(100):
        sample = df["Average_Spending"].sample(n=size)
        means.append(sample.mean())
    print("Sample Size:", size)
    print("Mean:", round(np.mean(means), 2))
    print("Standard Deviation:", round(np.std(means, ddof=1), 2))
    print()
```

OUTPUT :-

```
Sample Size: 10
Mean: 542.21
Standard Deviation: 48.49

Sample Size: 20
Mean: 533.82
Standard Deviation: 30.36

Sample Size: 30
Mean: 533.41
Standard Deviation: 22.62

Sample Size: 40
Mean: 533.14
Standard Deviation: 11.73
```